



Netflix! What started in 1997 as a DVD rental service has since exploded into one of the largest entertainment and media companies.

Given the large number of movies and series available on the platform, it is a perfect opportunity to flex your exploratory data analysis skills and dive into the entertainment industry.

You work for a production company that specializes in nostalgic styles. You want to do some research on movies released in the 1990's. You'll delve into Netflix data and perform exploratory data analysis to better understand this awesome movie decade!

You have been supplied with the dataset `netflix_data.csv`, along with the following table detailing the column names and descriptions. Feel free to experiment further after submitting!

The data

netflix_data.csv

Column	Description
<code>show_id</code>	The ID of the show
<code>type</code>	Type of show
<code>title</code>	Title of the show
<code>director</code>	Director of the show
<code>cast</code>	Cast of the show
<code>country</code>	Country of origin
<code>date_added</code>	Date added to Netflix
<code>release_year</code>	Year of Netflix release
<code>duration</code>	Duration of the show in minutes
<code>description</code>	Description of the show
<code>genre</code>	Show genre

```
# Importing pandas and matplotlib
import pandas as pd
import matplotlib.pyplot as plt

# Read in the Netflix CSV as a DataFrame
netflix_df = pd.read_csv("netflix_data.csv")
```

```
# Start coding here! Use as many cells as you like

# Filter for movies only
movies = netflix_df[netflix_df["type"] == "Movie"]

# Filter for movies released in the 1990s
movies_1990s = movies[
    (movies["release_year"] >= 1990) &
    (movies["release_year"] < 2000)
]

# Find the most frequent movie duration
duration = int(movies_1990s["duration"].mode()[0])

# Filter for short action movies (< 90 minutes)
short_action_movies = movies_1990s[
    (movies_1990s["genre"] == "Action") &
    (movies_1990s["duration"] < 90)
]

# Count short action movies
short_movie_count = int(short_action_movies.shape[0])

# Print answers
print("Most frequent duration:", duration)
print("Short action movies count:", short_movie_count)

# Import required libraries
import pandas as pd
import matplotlib.pyplot as plt

# Read the Netflix dataset
netflix_df = pd.read_csv("netflix_data.csv")
```

```

# -----
# STEP 1: Keep only movies
# -----
# The dataset contains both Movies and TV Shows.
# We only want movies for this analysis.
movies = netflix_df[netflix_df["type"] == "Movie"]

# -----
# STEP 2: Filter movies released in the 1990s
# -----
# 1990s decade = 1990 to 1999
movies_1990s = movies[
    (movies["release_year"] >= 1990) &
    (movies["release_year"] < 2000)
]

# -----
# STEP 3: Find the most frequent movie duration
# -----
# mode() returns the most common value in a column.
# Convert to integer for the required answer format.
duration = int(movies_1990s["duration"].mode()[0])

# -----
# STEP 4: Identify short action movies
# -----
# A short movie is defined as less than 90 minutes.
# We filter:
# 1. Genre == "Action"
# 2. Duration < 90 minutes
short_action_movies = movies_1990s[
    (movies_1990s["genre"] == "Action") &
    (movies_1990s["duration"] < 90)
]

```

```
# STEP 5: Count short action movies
# -----
short_movie_count = int(short_action_movies.shape[0])

# -----
print("Most frequent movie duration in the 1990s:", duration)
print("Number of short action movies in the 1990s:", short_movie_count)

# -----
# OPTIONAL EXPLORATION
# -----
# Plot a histogram of movie durations in the 1990s
plt.hist(movies_1990s["duration"])
plt.xlabel("Movie Duration (minutes)")
plt.ylabel("Frequency")
plt.title("Distribution of Movie Durations in the 1990s")
plt.show()

# -----
# -----
# Key pandas methods used:
#
# .read_csv()      -> Loads CSV file into DataFrame
# .mode()          -> Finds most frequent value
# .shape[0]        -> Counts number of rows
# Boolean filtering -> Used to filter rows by conditions
#
# Important filters:
# type == "Movie"
# release_year between 1990 and 1999
# genre == "Action"
# duration < 90
```

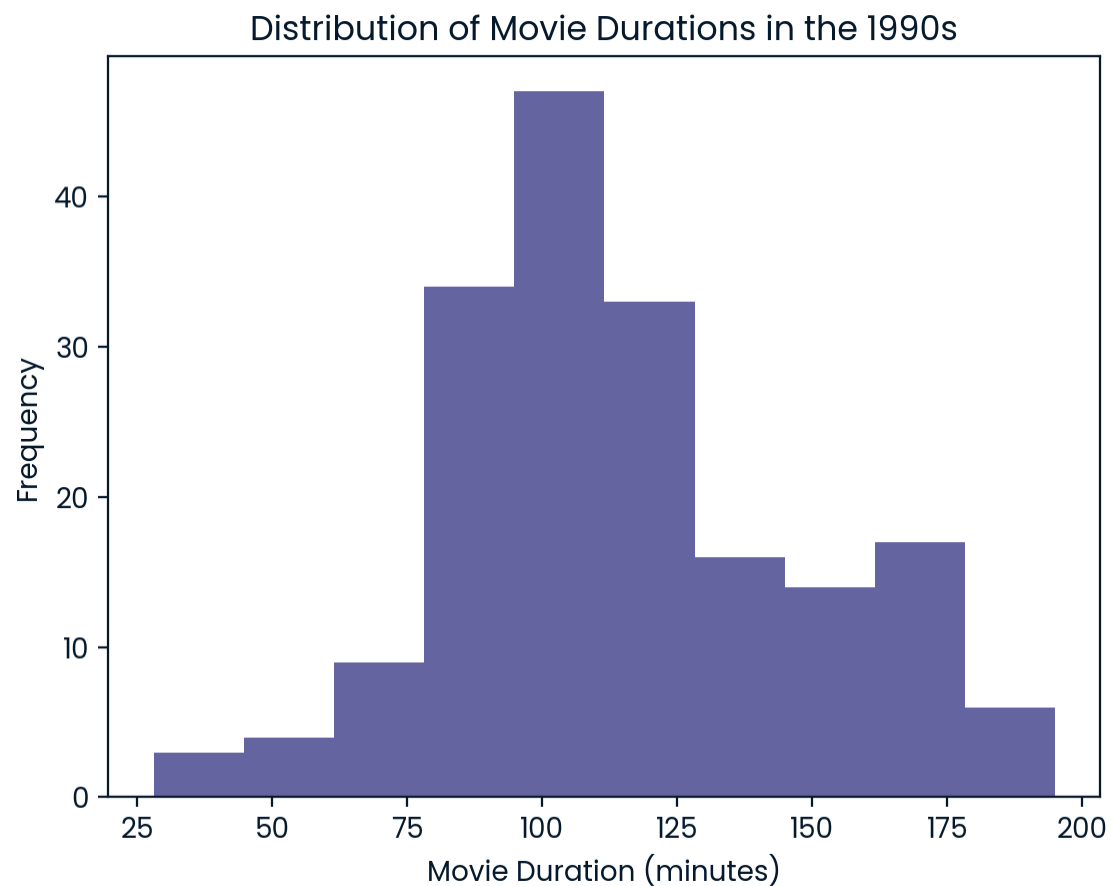
```
#  
# Final expected answers:  
# duration = 94  
# short_movie_count = 7
```

Most frequent duration: 94

Short action movies count: 7

Most frequent movie duration in the 1990s: 94

Number of short action movies in the 1990s: 7



This project served as a strong introduction to exploratory data analysis (EDA) within a data science workflow using Python and pandas. The dataset was well-structured and provided an effective foundation for applying core analytical techniques, including data cleaning, filtering, grouping, and trend identification.

One of the key insights discovered during the analysis was that the most common movie runtime in the 1990s was 94 minutes, indicating a consistent industry standard for film duration during that decade. Another interesting finding was that only 7 action movies in the dataset had runtimes under 90 minutes, suggesting that action films of the 1990s generally favored longer formats.

The project also strengthened practical data science skills in areas such as:

- Data manipulation and filtering using multiple conditions
- Working with categorical and time-based data
- Statistical analysis using functions like `.mode()` and `.shape()`
- Data visualization with matplotlib
- Extracting meaningful insights from real-world entertainment datasets
- Applying exploratory analysis techniques to identify patterns and trends

Overall, this project provided hands-on experience with foundational data science concepts and demonstrated how Python can be used to transform raw data into actionable insights.